

Integrated Circuit ECE326

Lec. 08

Synthesis Digital Circuit

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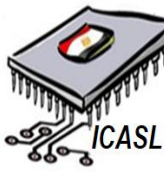


- ☐ Transistor Sizing
- ☐ Size of CMOS digital circuit
- ☐ Synthesize and Size of CMOS digital circuit

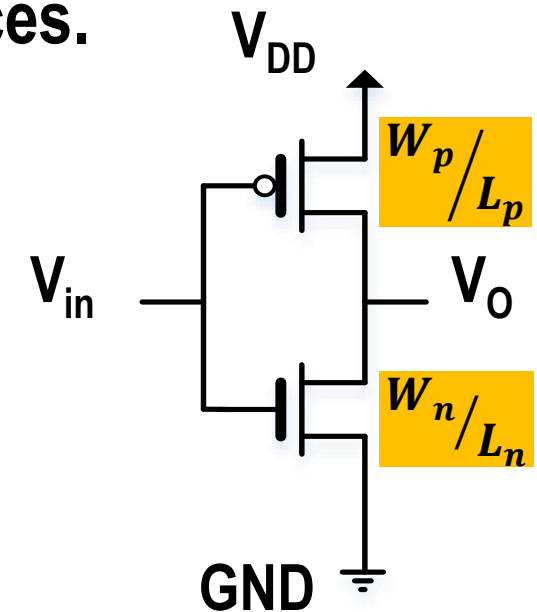


- ❑ **Transistor Sizing**
- ❑ **Size of CMOS digital circuit**
- ❑ **Synthesize and Size of CMOS digital circuit**

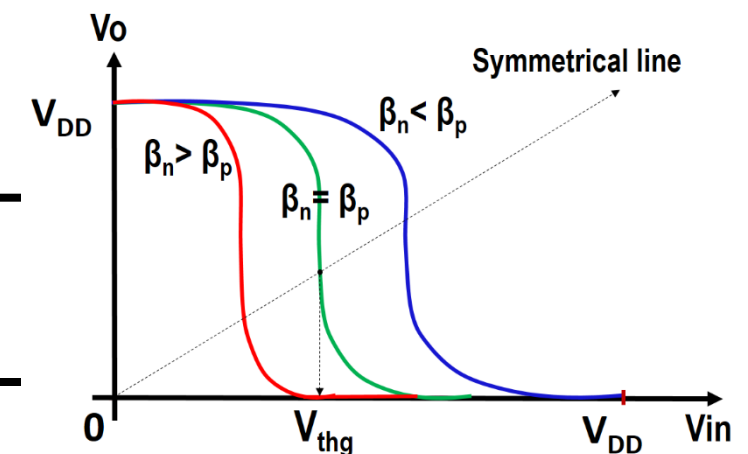
Transistor Sizing



- Once a CMOS gate circuit has been **synthesized**, the next significant step remaining in the design is to decide **W/L** ratios for all devices.



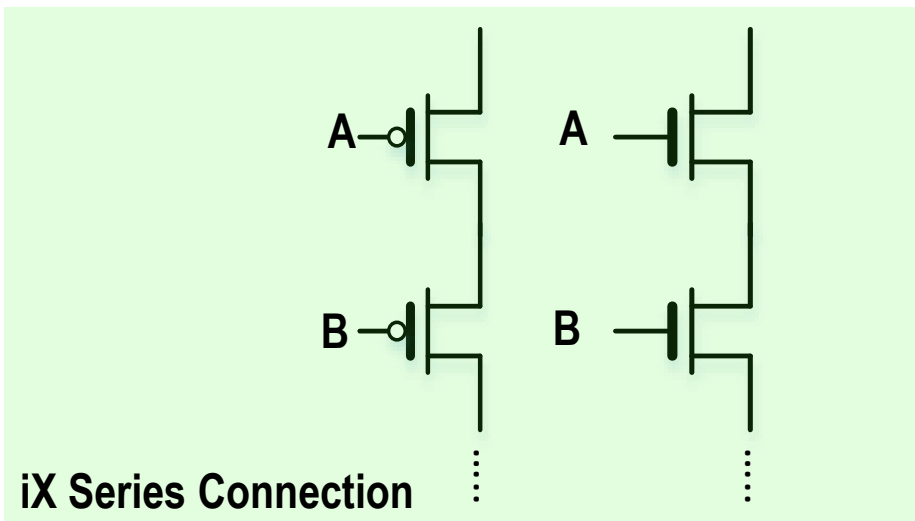
Minimum Area	Symmetrical condition (Matched design)
$W_p = W_n \text{ \& } L_p = L_n$  $\beta_n \neq \beta_p$	$W_p = 2.5 W_n \text{ \& } L_p = L_n$  $\beta_n = \beta_p$
Assume $L_p = L_n = 1\mu\text{m}$ $W_n = 2\mu\text{m} \rightarrow W_p = 2\mu\text{m}$	Assume $L_p = L_n = 1\mu\text{m}$ $W_n = 2\mu\text{m} \rightarrow W_p = 5\mu\text{m}$



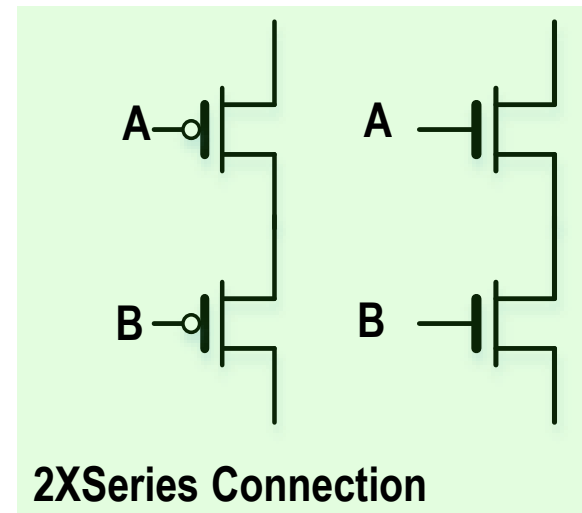
Transistor Sizing



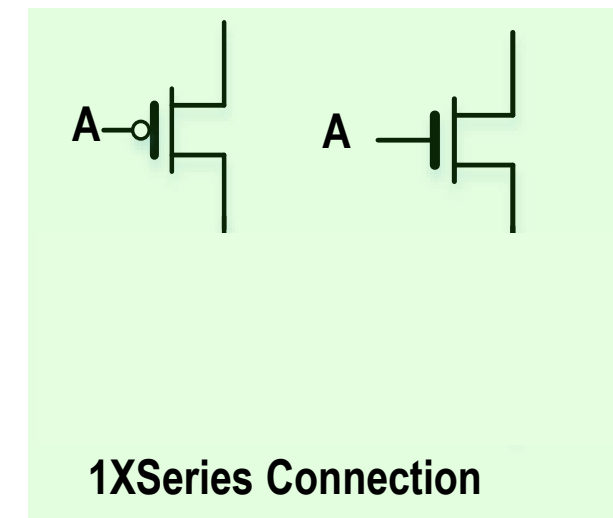
- ❑ Steps for transistor sizing of the digital circuit
 - ❑ **1st Step:** Synthesize the required digital function.
 - ❑ **2nd Step:** Design the reference (Equivalent) CMOS inverter
 - ❑ **3rd Step:** Distribute size of the reference inverter over the worst path of Synthesized digital circuit.



$$\frac{1}{(W/L)_{eq.}} = \frac{1}{(W/L)_A} + \frac{1}{(W/L)_B} + \dots + \frac{1}{(W/L)_i}$$



$$\frac{1}{(W/L)_{eq.}} = \frac{1}{(W/L)_A} + \frac{1}{(W/L)_B}$$



$$(W/L)_{eq.} = (W/L)_A$$



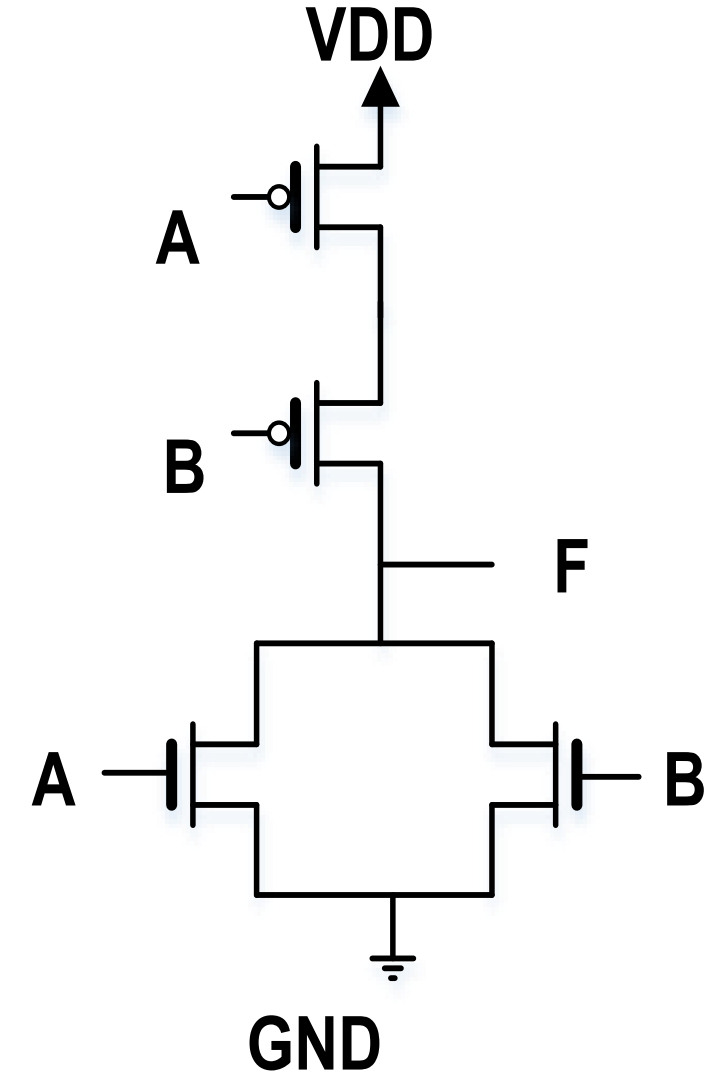
- ❑ Transistor Sizing
- ❑ Size of CMOS digital circuit
- ❑ Synthesize and Size of CMOS digital circuit

Size of CMOS digital circuit (1/15)



Ex.1 Sizing the following circuit using a matched design of a reference inverter .

Ans



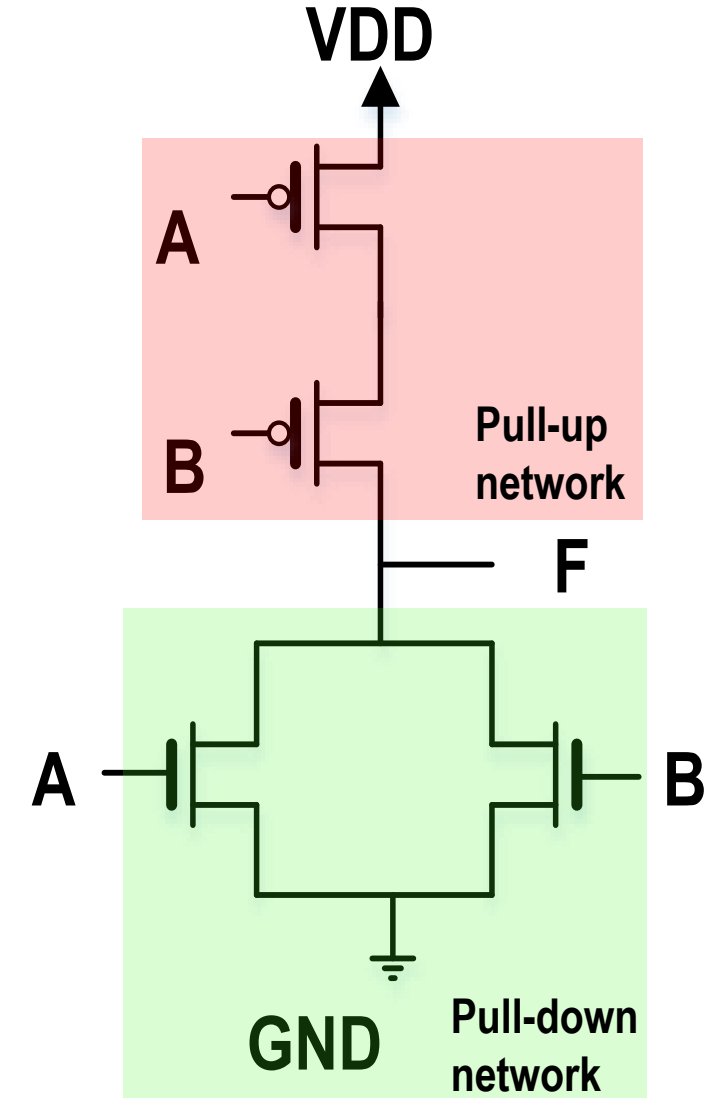
Size of CMOS digital circuit (2/15)



Ex.1 Sizing the following circuit using a matched design of a reference inverter .

Ans

1st Step: Synthesize the required function (**Given**)



Size of CMOS digital circuit (3/15)

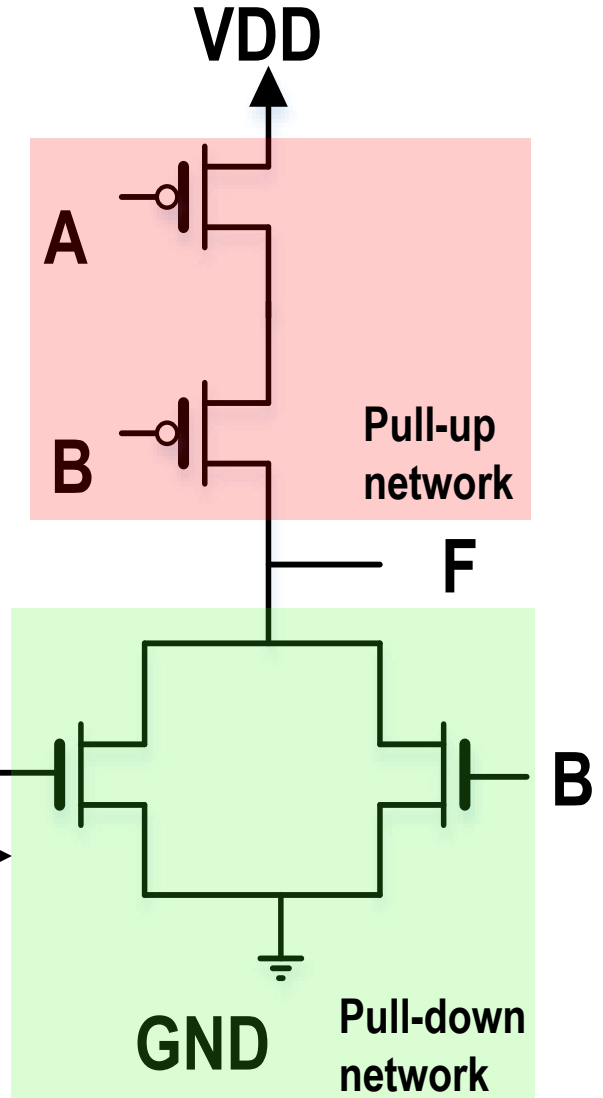
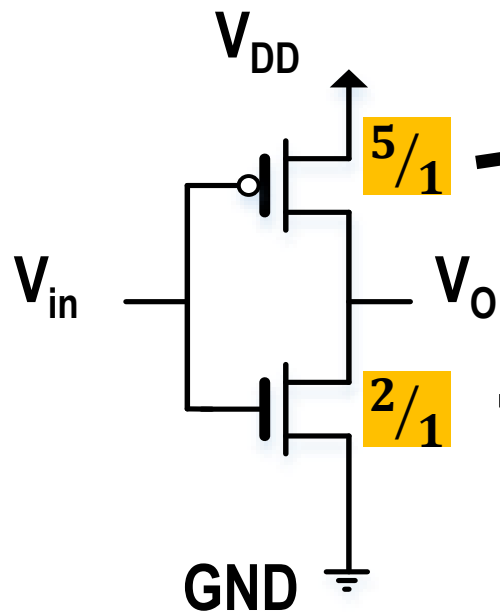
Ex.1 Sizing the following circuit using a matched design of a reference inverter .

Ans

2nd Step: Design the reference (Equivalent) CMOS inverter

Assume $L_p = L_n = 1\mu\text{m}$
 $W_n = 2\mu\text{m} \rightarrow W_p = 5\mu\text{m}$

$\beta_n = \beta_p$ Symmetrical condition
 (Matched design)



$$(W/L)_{\text{Ref.}} = (W/L)_{\text{eq.}}$$

Size of CMOS digital circuit (4/15)



Ex.1 Sizing the following circuit using a matched design of a reference inverter .

Ans

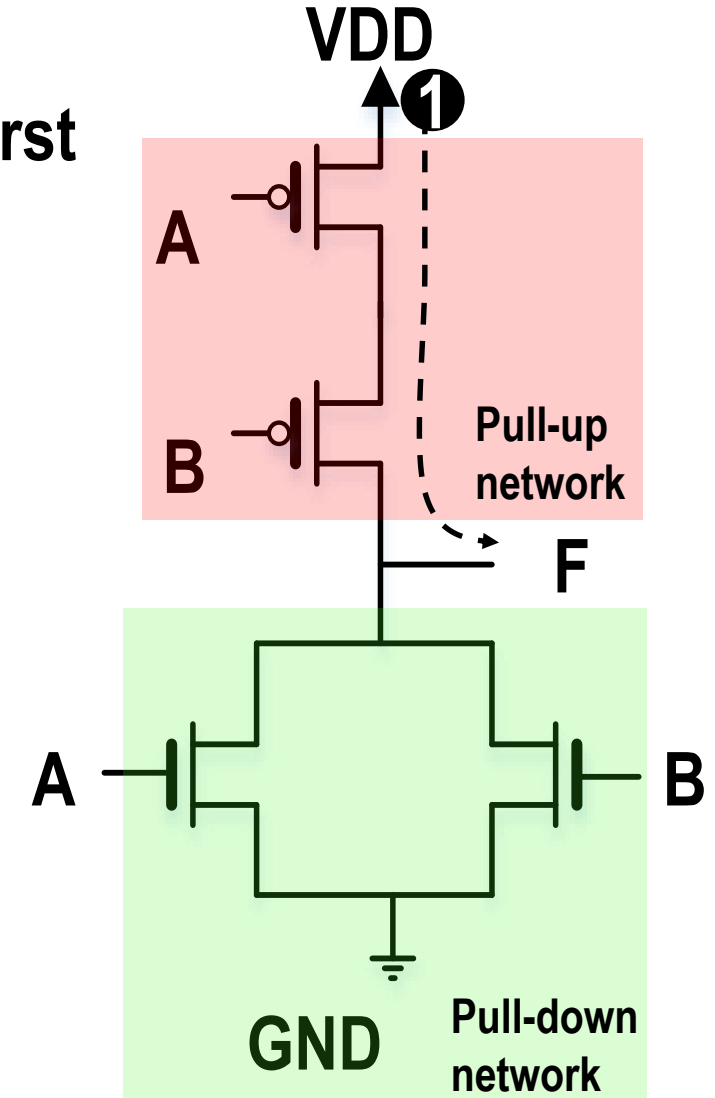
3rd Step: Distribute size of the reference inverter over the worst path of required digital circuit.

□ **For Pull-Up network (Worst Path_1)**

$$\frac{1}{(W/L)_{eq.}} = \frac{1}{(W/L)_A} + \frac{1}{(W/L)_B} \quad \xrightarrow{\text{Assume}} \quad (W/L)_A = (W/L)_B$$

$$\frac{1}{(W/L)_{eq.}} = \frac{2}{(W/L)_{A,B}}$$

$$\frac{1}{5} = \frac{2}{(W/L)_{A,B}} \quad \xrightarrow{\quad} \quad (W/L)_A = (W/L)_B = 10/1$$



Size of CMOS digital circuit (5/15)



Ex.1 Sizing the following circuit using a matched design of a reference inverter .

Ans

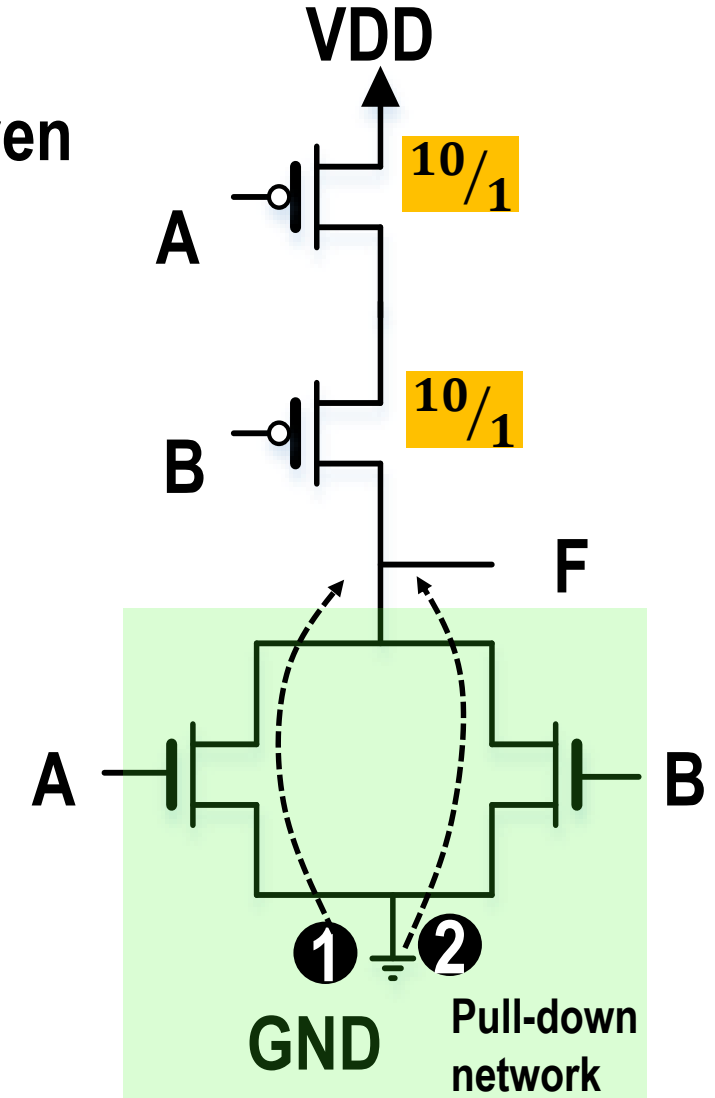
3rd Step: Distribute size of the reference inverter over the given worst path of required digital circuit.

□ For Pull-down network (**Worst Path_1**)

$$(W/L)_{eq.} = (W/L)_A = 2/1$$

□ For Pull-down network (**Worst Path_2**)

$$(W/L)_{eq.} = (W/L)_B = 2/1$$

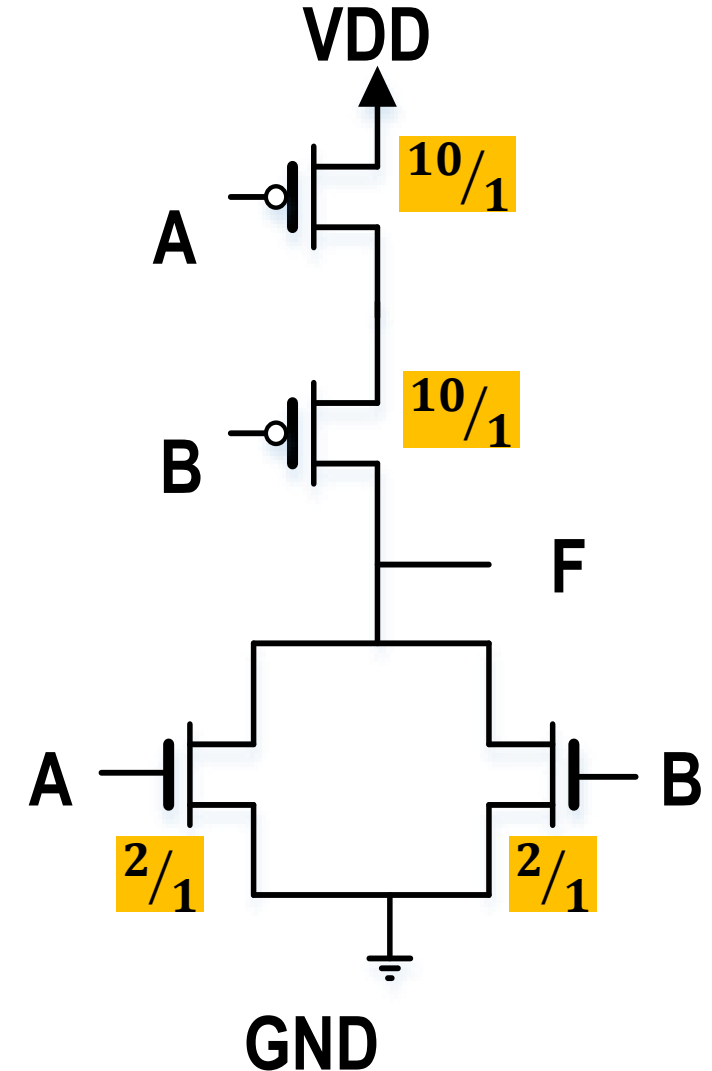
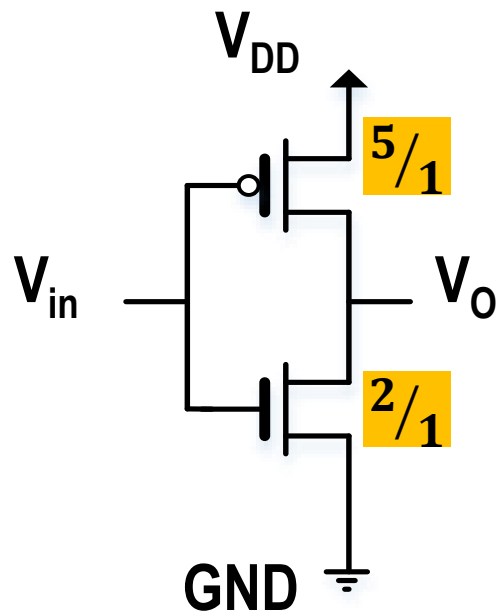


Size of CMOS digital circuit (6/15)



Ex.1 Sizing the following circuit using a matched design of a reference inverter .

Ans

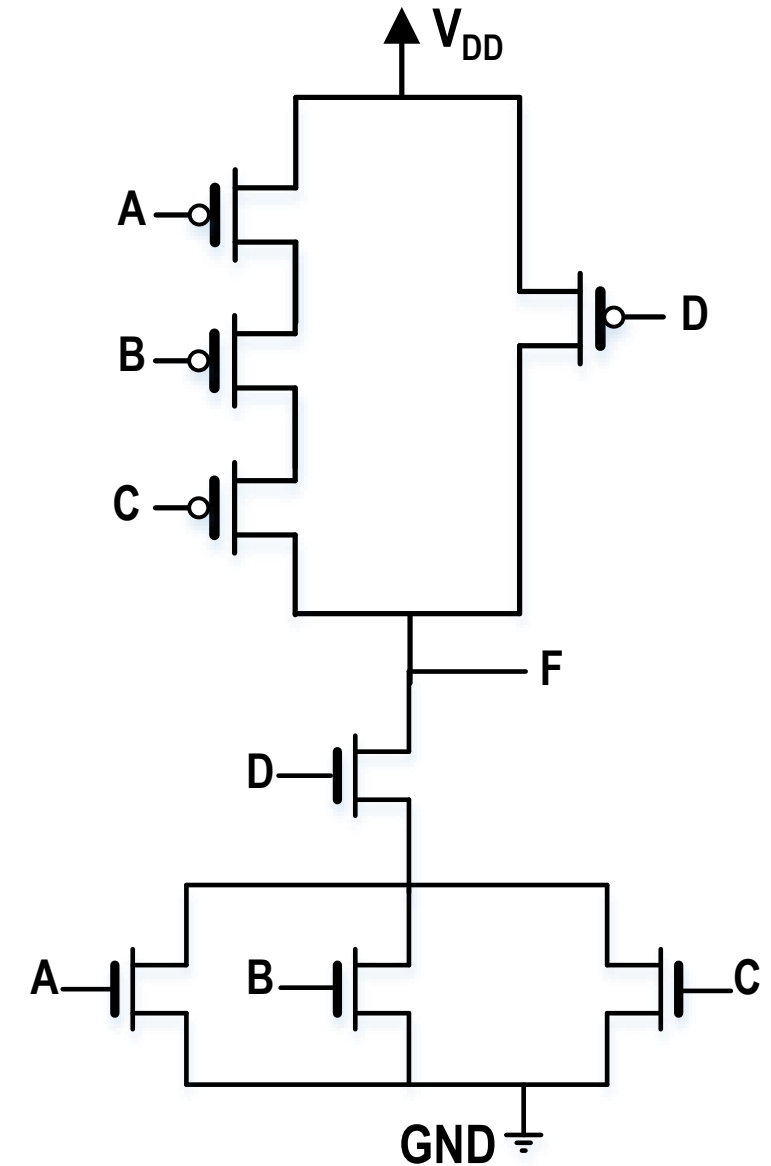


Size of CMOS digital circuit (7/15)



Ex.1 Sizing the following circuit using a matched design of a reference inverter .

Ans



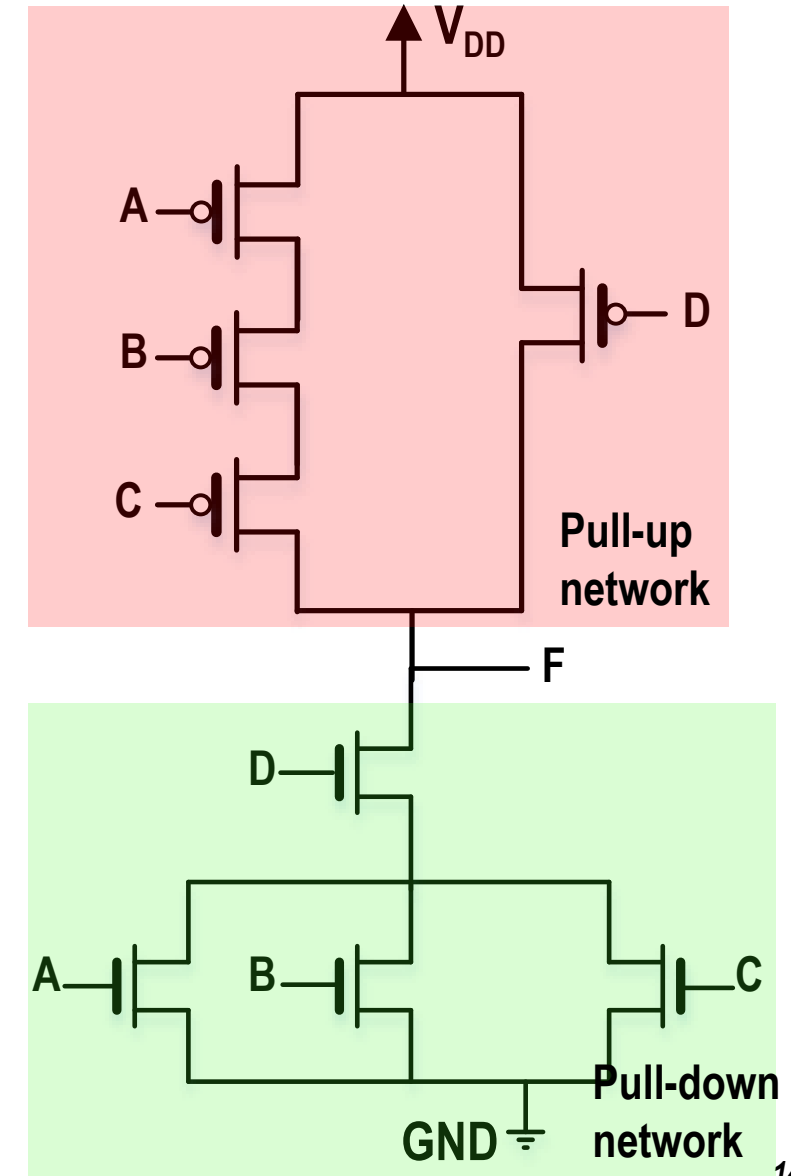
Size of CMOS digital circuit (8/15)



Ex.1 Sizing the following circuit using a matched design of a reference inverter .

Ans

1st Step: Synthesize the required function (**Given**)



Size of CMOS digital circuit (9/15)

Ex.1 Sizing the following circuit using a matched design of a reference inverter .

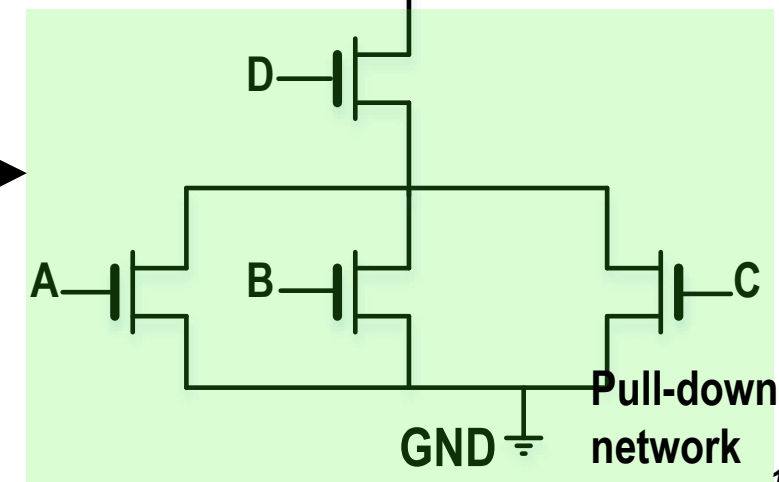
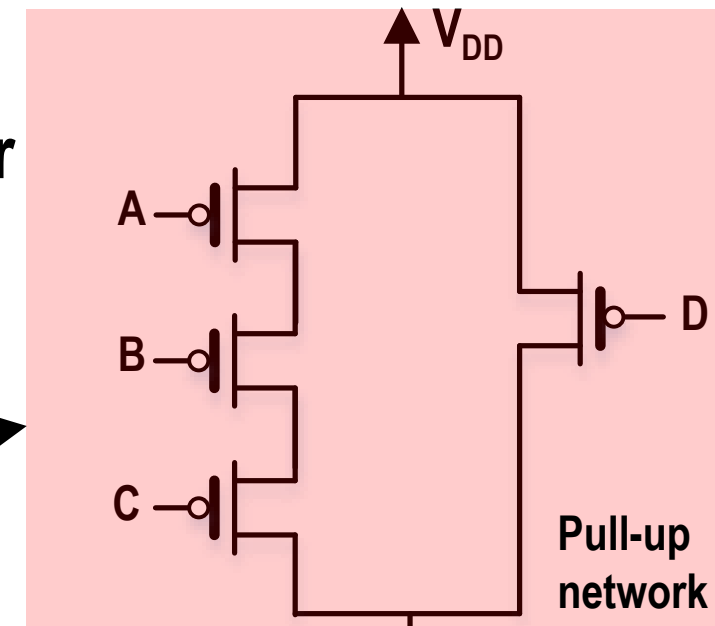
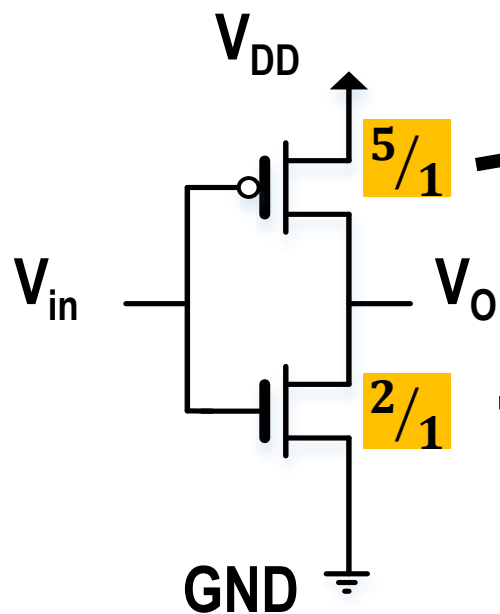
Ans

2nd Step: Design the reference (Equivalent) CMOS inverter

Assume $L_p = L_n = 1\mu\text{m}$
 $W_n = 2\mu\text{m} \rightarrow W_p = 5\mu\text{m}$



$\beta_n = \beta_p$ Symmetrical condition
 (Matched design)



$$(W/L)_{\text{Ref.}} = (W/L)_{\text{eq.}}$$

Size of CMOS digital circuit (10/15)

Ex.1 Sizing the following circuit using a matched design of a reference inverter .

Ans

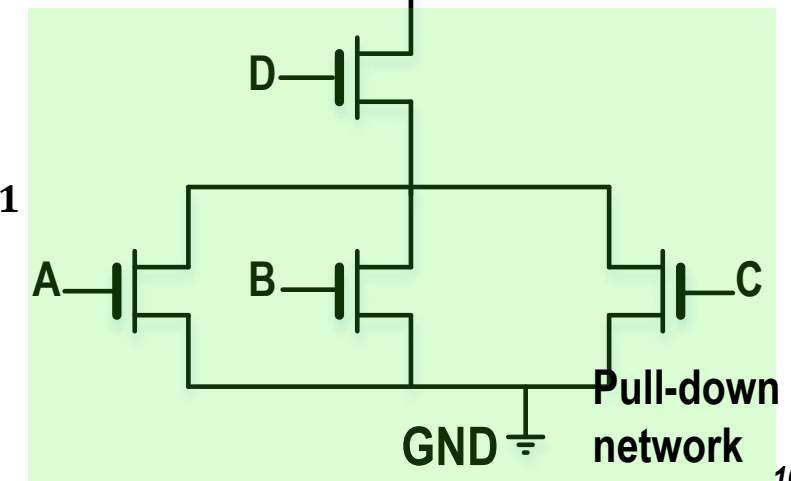
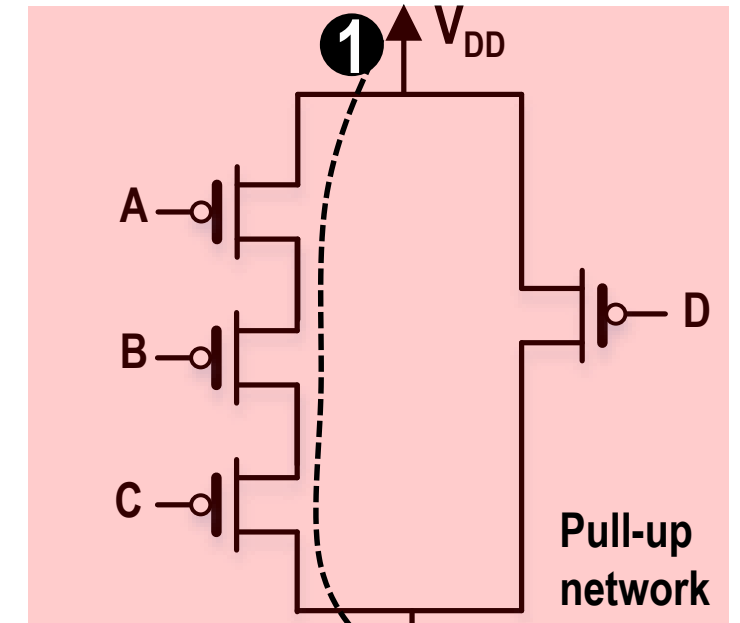
3rd Step: Distribute size of the reference inverter over the worst path of required digital circuit.

□ For Pull-Up network (Worst Path_1)

$$\frac{1}{(W/L)_{eq.}} = \frac{1}{(W/L)_A} + \frac{1}{(W/L)_B} + \frac{1}{(W/L)_C} \quad \text{Assume} \quad (W/L)_A = (W/L)_B = (W/L)_C$$

$$\frac{1}{(W/L)_{eq.}} = \frac{3}{(W/L)_{A,B,C}}$$

$$\frac{1}{5} = \frac{3}{(W/L)_{A,B,C}} \quad \Rightarrow \quad (W/L)_A = (W/L)_B = (W/L)_C = 15/1$$



Size of CMOS digital circuit (11/15)



Ex.1 Sizing the following circuit using a matched design of a reference inverter .

Ans

3rd Step: Distribute size of the reference inverter over the worst path of required digital circuit.

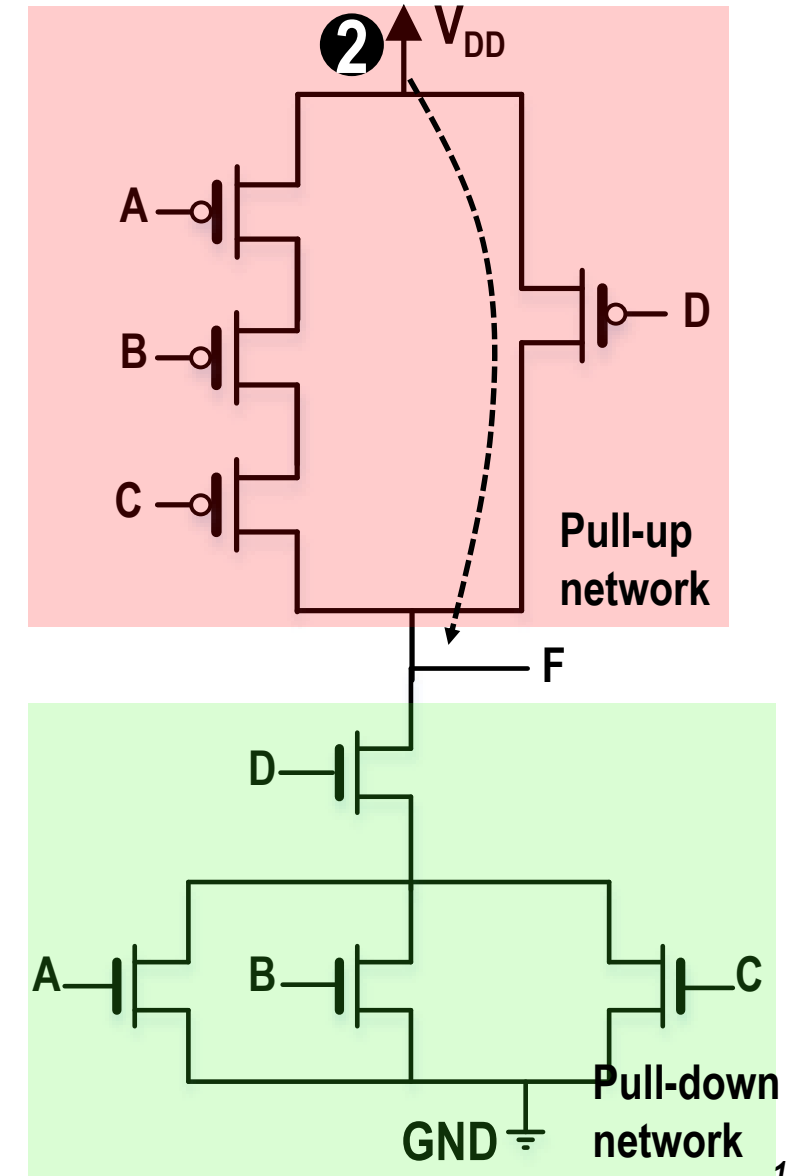
□ For Pull-Up network (Path_2)

$$\frac{1}{(W/L)_{eq.}} = \frac{1}{(W/L)_D}$$

$$\frac{1}{(W/L)_{eq.}} = \frac{1}{(W/L)_D}$$

$$\frac{1}{5} = \frac{1}{(W/L)_D}$$

→ $(W/L)_D = 5/1$



Size of CMOS digital circuit (12/15)



Ex.1 Sizing the following circuit using a matched design of a reference inverter .

Ans

3rd Step: Distribute size of the reference inverter over the worst path of required digital circuit.

□ For Pull-down network (Worst Path_1)

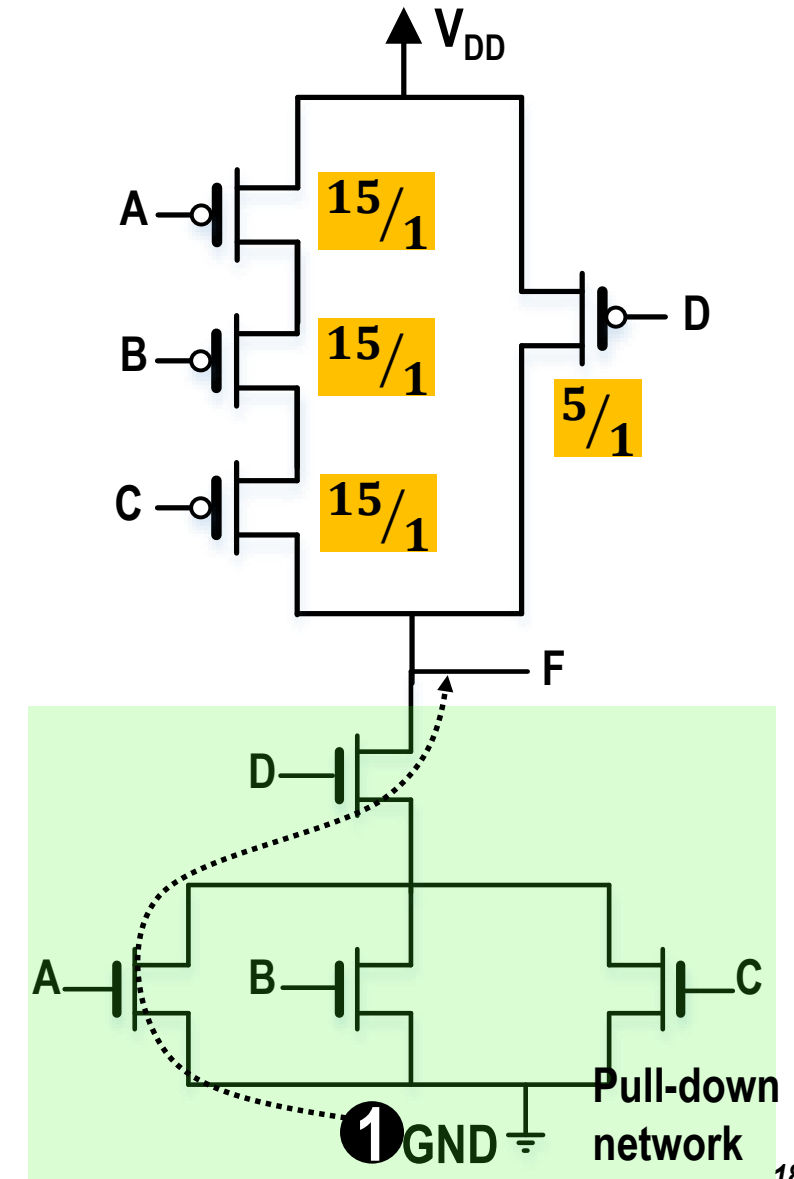
$$\frac{1}{(W/L)_{eq}} = \frac{1}{(W/L)_A} + \frac{1}{(W/L)_D}$$

Assume $(W/L)_A = (W/L)_D$

$$\frac{1}{(W/L)_{eq.}} = \frac{2}{(W/L)_{A,D}}$$

$$\frac{1}{2} = \frac{2}{(W/L)_{A,D}}$$

$(W/L)_A = (W/L)_D = 4/1$



Size of CMOS digital circuit (13/15)

Ex.1 Sizing the following circuit using a matched design of a reference inverter .

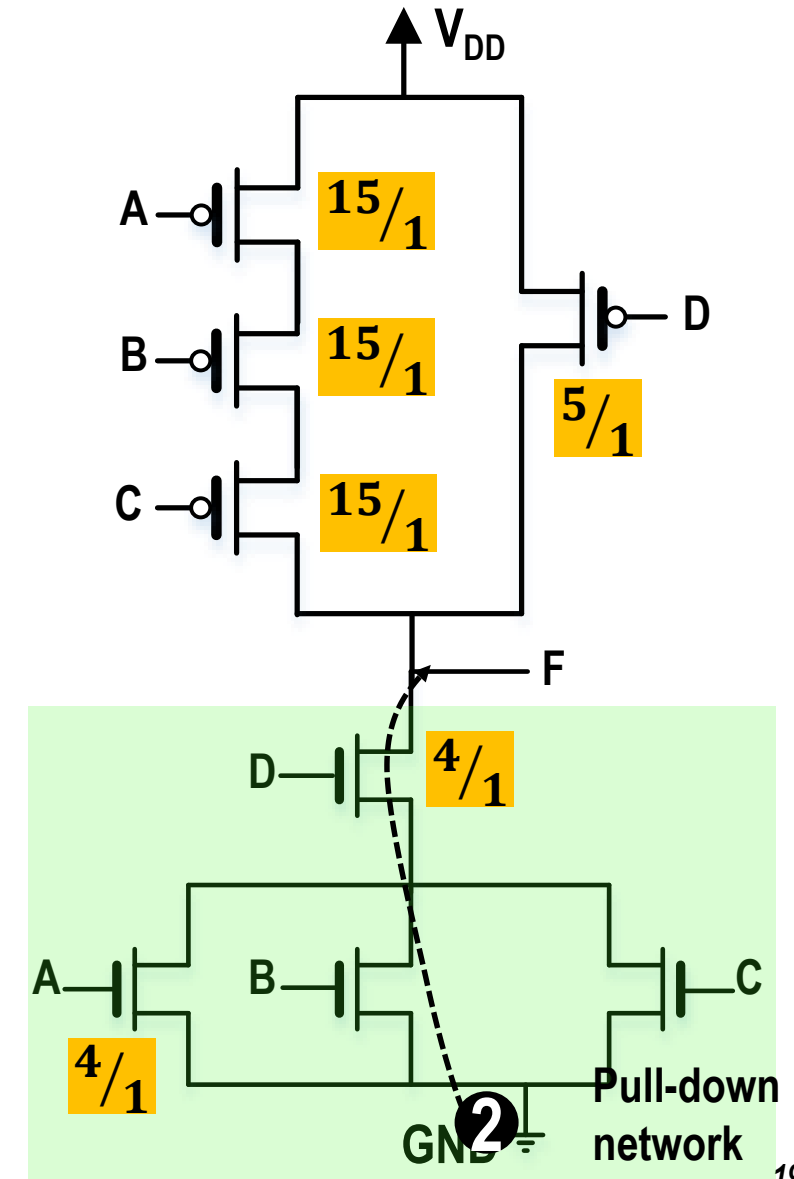
Ans

3rd Step: Distribute size of the reference inverter over the worst path of required digital circuit.

□ **For Pull-down network (Worst Path_2)**

$$\frac{1}{(W/L)_{eq.}} = \frac{1}{(W/L)_B} + \frac{1}{(W/L)_D}$$

$$\frac{1}{2} = \frac{1}{(W/L)_B} + \frac{1}{4} \quad \Rightarrow \quad (W/L)_B = 4/1$$



Size of CMOS digital circuit (14/15)

Ex.1 Sizing the following circuit using a matched design of a reference inverter .

Ans

3rd Step: Distribute size of the reference inverter over the worst path of required digital circuit.

□ **For Pull-down network (Worst Path_2)**

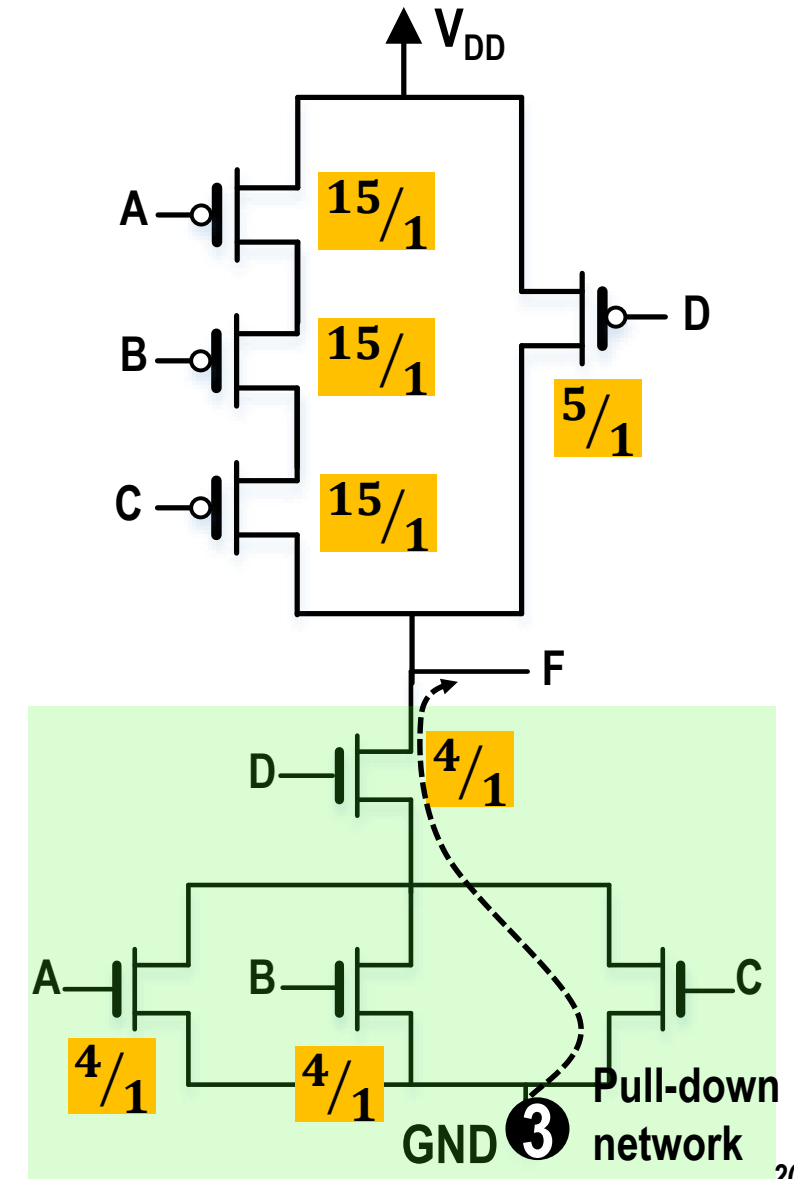
$$\frac{1}{(W/L)_{eq.}} = \frac{1}{(W/L)_B} + \frac{1}{(W/L)_D}$$

$$\frac{1}{2} = \frac{1}{(W/L)_B} + \frac{1}{4} \quad \Rightarrow \quad (W/L)_B = 4/1$$

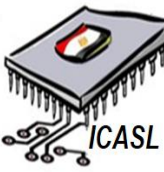
□ **For Pull-down network (Worst Path_3)**

$$\frac{1}{(W/L)_{eq.}} = \frac{1}{(W/L)_C} + \frac{1}{(W/L)_D}$$

$$\frac{1}{2} = \frac{1}{(W/L)_C} + \frac{1}{4} \quad \Rightarrow \quad (W/L)_C = 4/1$$

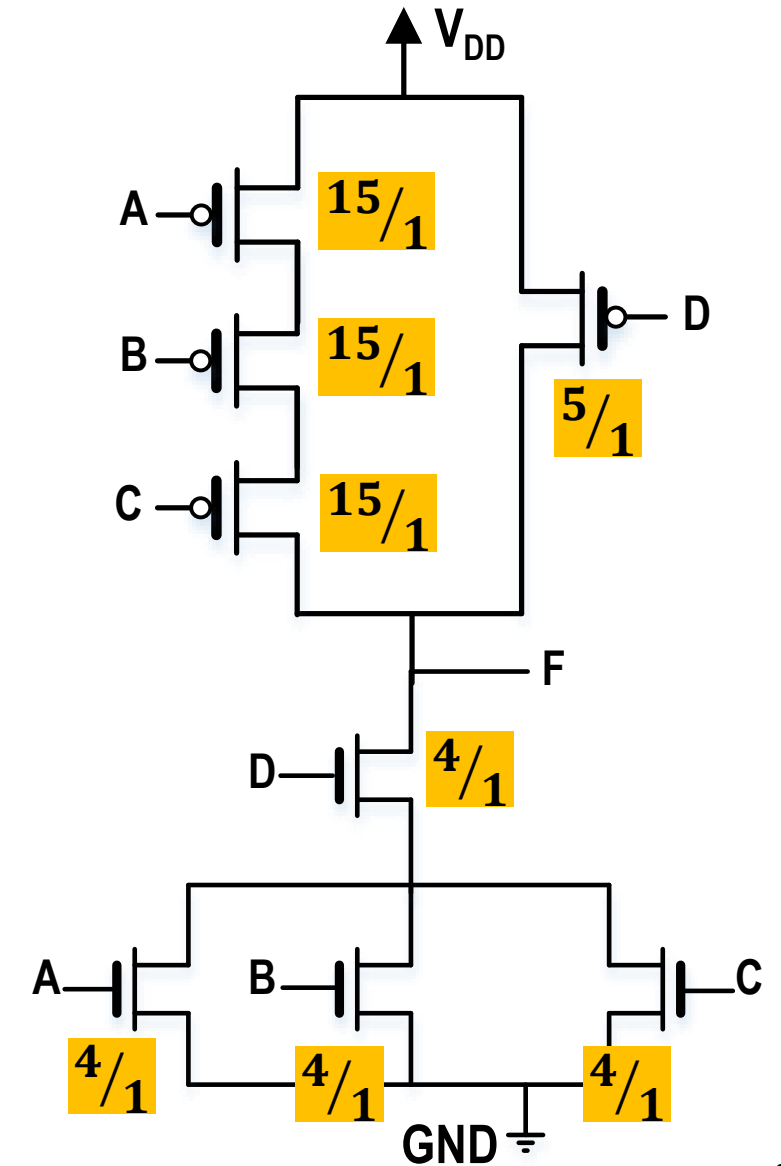
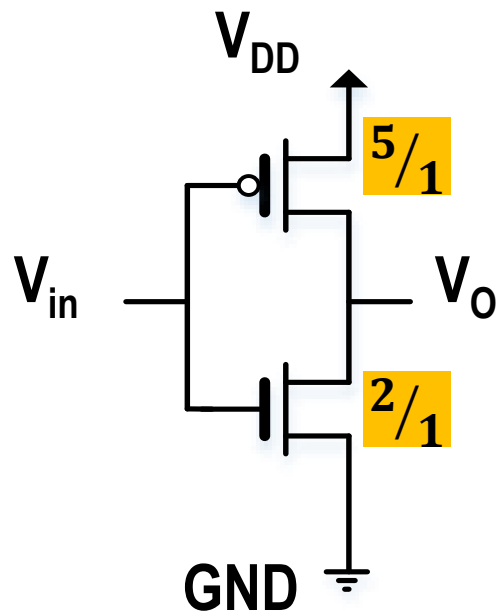


Size of CMOS digital circuit (15/15)



Ex.1 Sizing the following circuit using a matched design of a reference inverter .

Ans





- ❑ Transistor Sizing
- ❑ Size of CMOS digital circuit
- ❑ Synthesize and Size of CMOS digital circuit

Synthesize and Size of CMOS digital circuit (1/10)



Ex.2. (a) Synthesize the function $F = \overline{A + B(C + D)}$ using CMOS transistors, (b) Sizing the synthesized circuit using a matched design of a reference inverter.

Ans.

1st step: Synthesize the required function

Step 1: $F = \overline{A + B(C + D)}$ is simplified directly

Step 2: $\overline{F} = A + B(C + D)$

You don not
need to use
K-Map

Synthesize and Size of CMOS digital circuit (2/10)



Ex.2. (a) Synthesize the function $F = \overline{A + B(C + D)}$ using CMOS transistors, (b) Sizing the synthesized circuit using a matched design of a reference inverter.

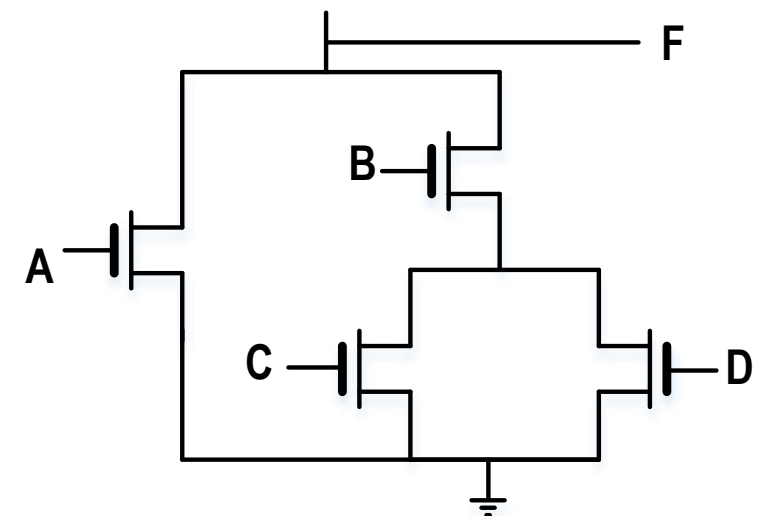
Ans.

1st step: Synthesize the required function

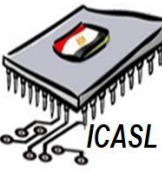
Step 2: $\overline{F} = A + B(C + D)$

Step 3: **Firstly**, draw the **pull-down** network

- ❑ Two transistors {C, D} are connected in parallel connection
- ❑ Transistors {C, D} are connected **with** {B} in series connection.
- ❑ Then all drawn transistors are connected with transistor {A} in parallel connection



Synthesize and Size of CMOS digital circuit (3/10)



Ex.2. (a) Synthesize the function $F = \overline{A + B(C + D)}$ using CMOS transistors, (b) Sizing the synthesized circuit using a matched design of a reference inverter.

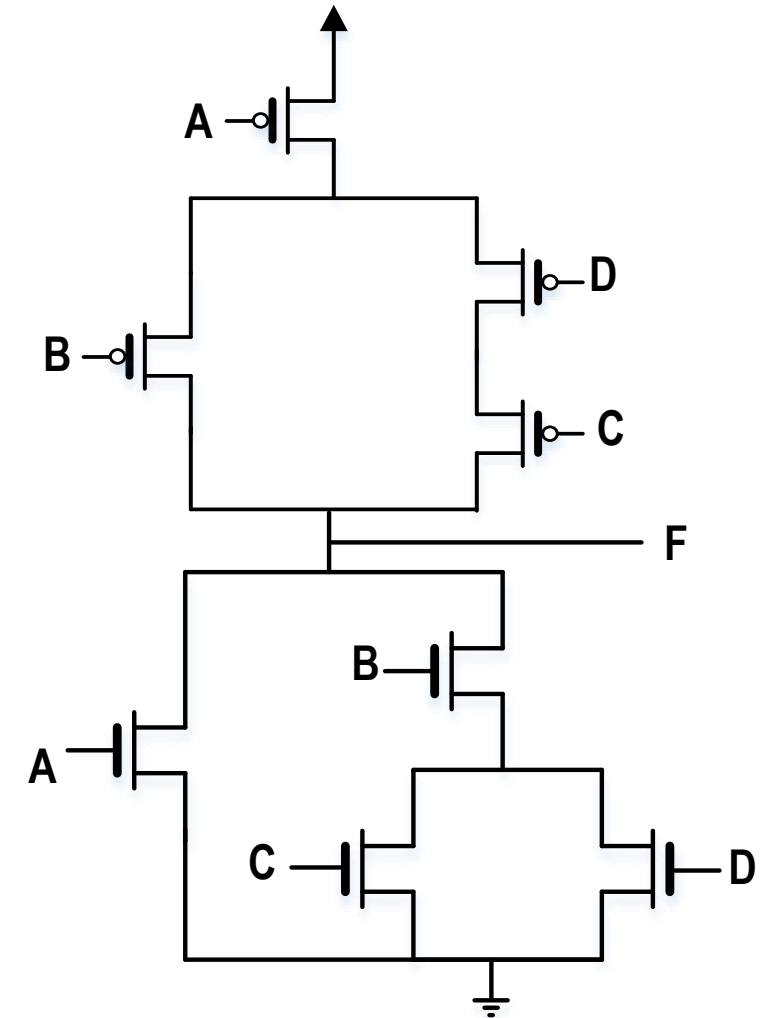
Ans.

1st step: Synthesize the required function

Step 2: $\overline{F} = A + B(C + D)$

Step 4: Secondly, draw the pull-up network

- ❑ Two transistors {C, D} are connected in series connection
- ❑ Transistors {C, D} are connected **with** {B} in parallel connection.
- ❑ Then all drawn transistors are connected with transistor {A} in series connection



Synthesize and Size of CMOS digital circuit (4/10)



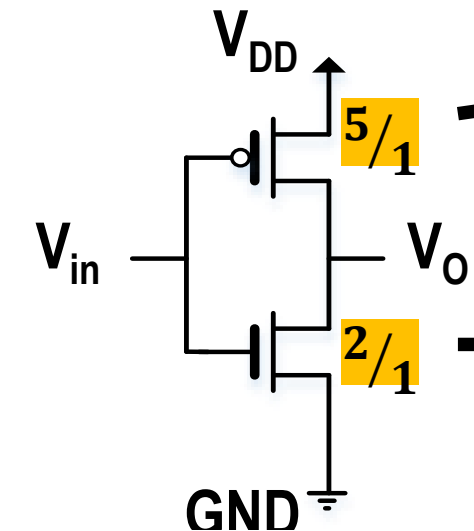
Ex.2. (a) Synthesize the function $F = \overline{A + B(C + D)}$ using CMOS transistors, (b) Sizing the synthesized circuit using a matched design of a reference inverter.

Ans.

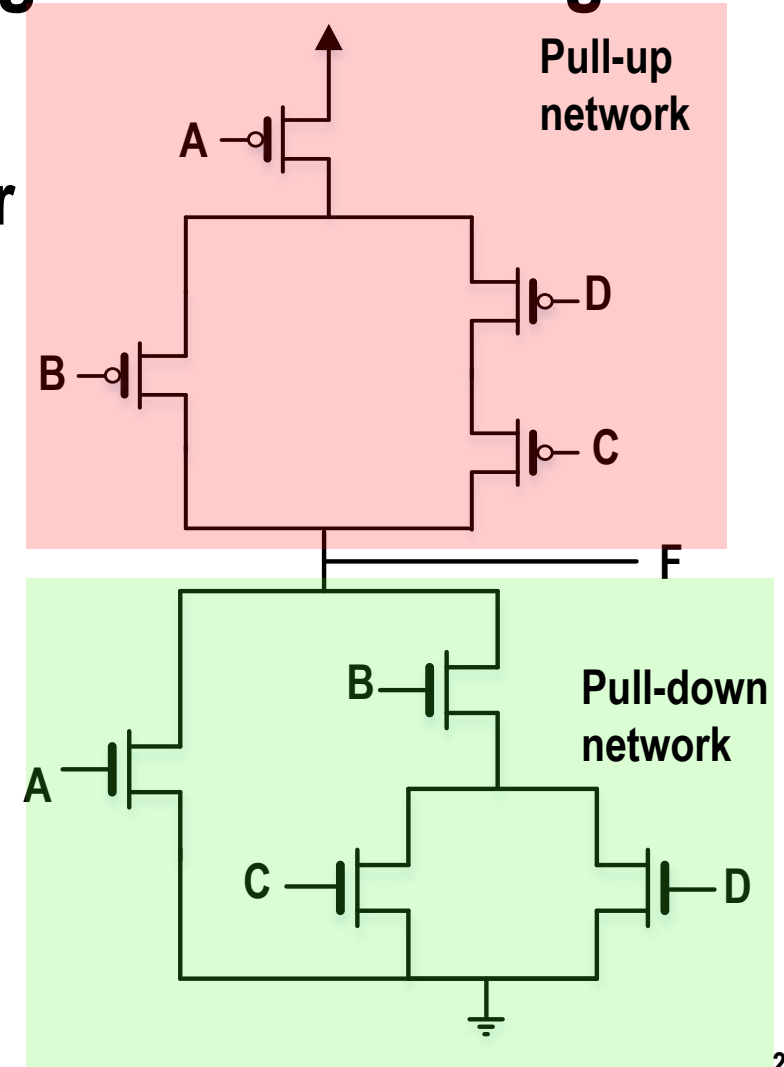
2nd Step: Design the reference (Equivalent) CMOS inverter

Assume $L_p = L_n = 1\mu\text{m}$
 $W_n = 2\mu\text{m} \rightarrow W_p = 5\mu\text{m}$

$\beta_n = \beta_p$ Symmetrical condition
(Matched design)



$$(W/L)_{\text{Ref.}} = (W/L)_{\text{eq.}}$$



Synthesize and Size of CMOS digital circuit (5/10)



Ex.2. (a) Synthesize the function $F = \overline{A + B(C + D)}$ using CMOS transistors, (b) Sizing the synthesized circuit using a matched design of a reference inverter.

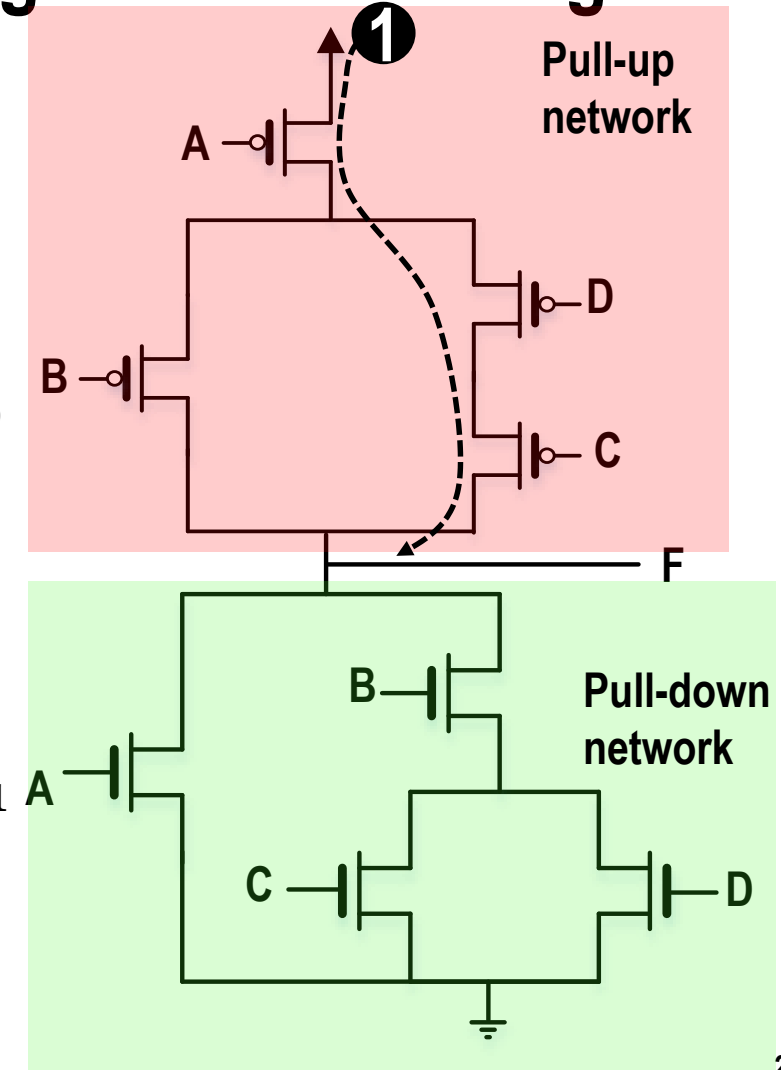
Ans.

□ For Pull-Up network (Worst Path_1)

$$\frac{1}{(W/L)_{eq.}} = \frac{1}{(W/L)_A} + \frac{1}{(W/L)_C} + \frac{1}{(W/L)_D} \quad \text{Assume} \quad (W/L)_A = (W/L)_C = (W/L)_D$$

$$\frac{1}{(W/L)_{Ref.}} = \frac{3}{(W/L)_{A,C,D}}$$

$$\frac{1}{5} = \frac{3}{(W/L)_{A,C,D}} \quad \Rightarrow \quad (W/L)_A = (W/L)_C = (W/L)_D = 15/1$$



Synthesize and Size of CMOS digital circuit (6/10)



Ex.2. (a) Synthesize the function $F = \overline{A + B(C + D)}$ using CMOS transistors, (b) Sizing the synthesized circuit using a matched design of a reference inverter.

Ans.

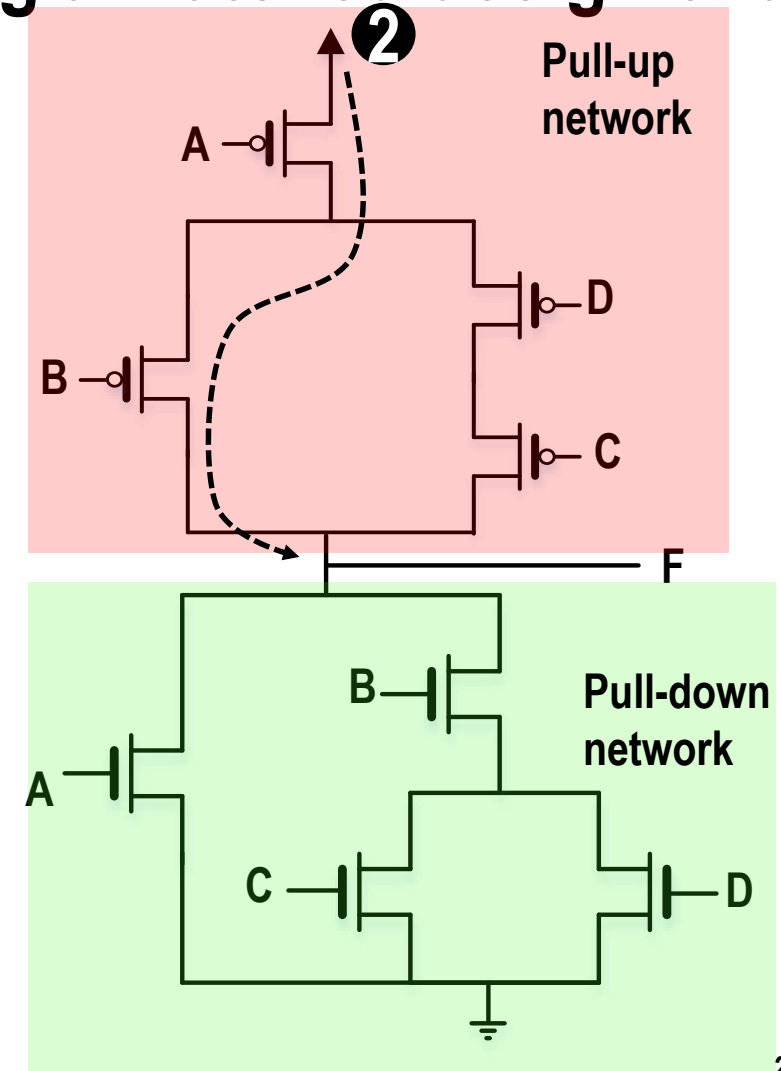
□ For Pull-Up network (Path_2)

$$\frac{1}{(W/L)_{eq.}} = \frac{1}{(W/L)_A} + \frac{1}{(W/L)_B}$$

$$\frac{1}{5} = \frac{1}{15} + \frac{1}{(W/L)_B}$$

$$\frac{1}{7.5} = \frac{1}{(W/L)_B}$$

➔ $(W/L)_B = 7.5/1$



Synthesize and Size of CMOS digital circuit (7/10)



Ex.2. (a) Synthesize the function $F = \overline{A + B(C + D)}$ using CMOS transistors, (b) Sizing the synthesized circuit using a matched design of a reference inverter.

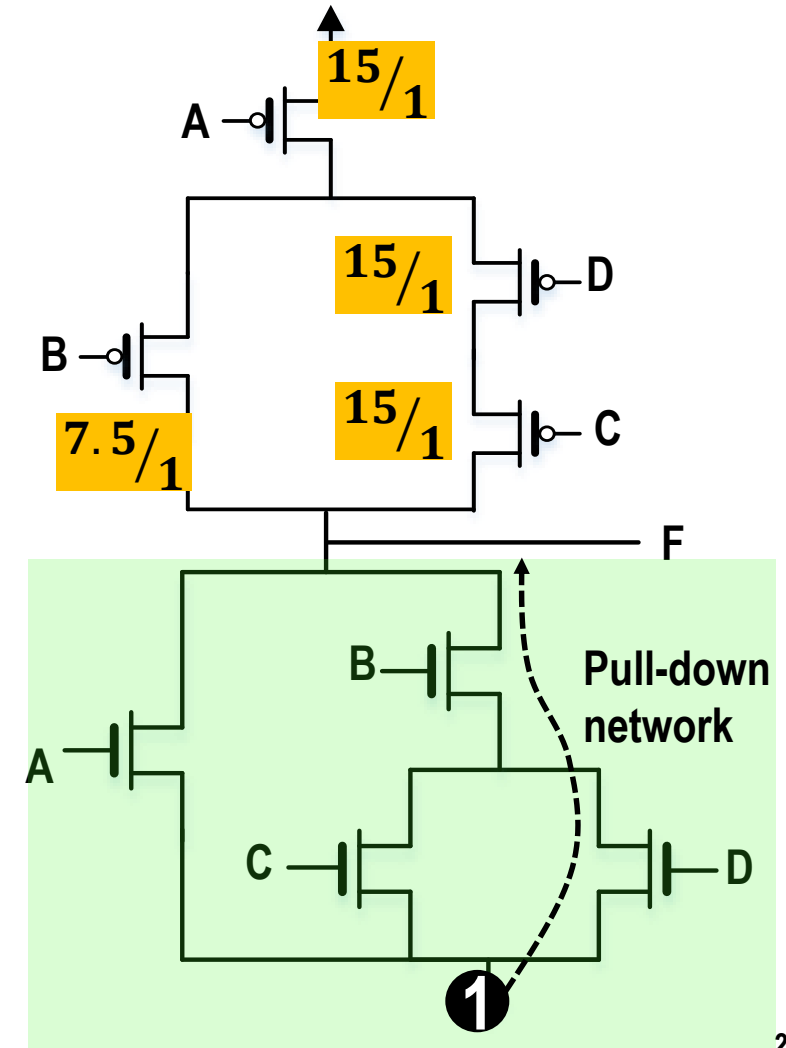
Ans.

□ For Pull-down network (**Worst Path_1**)

$$\frac{1}{(W/L)_{eq.}} = \frac{1}{(W/L)_B} + \frac{1}{(W/L)_D} \quad \xrightarrow{\text{Assume}} \quad (W/L)_B = (W/L)_D$$

$$\frac{1}{(W/L)_{eq.}} = \frac{2}{(W/L)_{B,D}}$$

$$\frac{1}{2} = \frac{2}{(W/L)_{B,D}} \quad \xrightarrow{\quad} \quad (W/L)_B = (W/L)_D = 4/1$$



Synthesize and Size of CMOS digital circuit (8/10)



Ex.2. (a) Synthesize the function $F = \overline{A + B(C + D)}$ using CMOS transistors, (b) Sizing the synthesized circuit using a matched design of a reference inverter.

Ans.

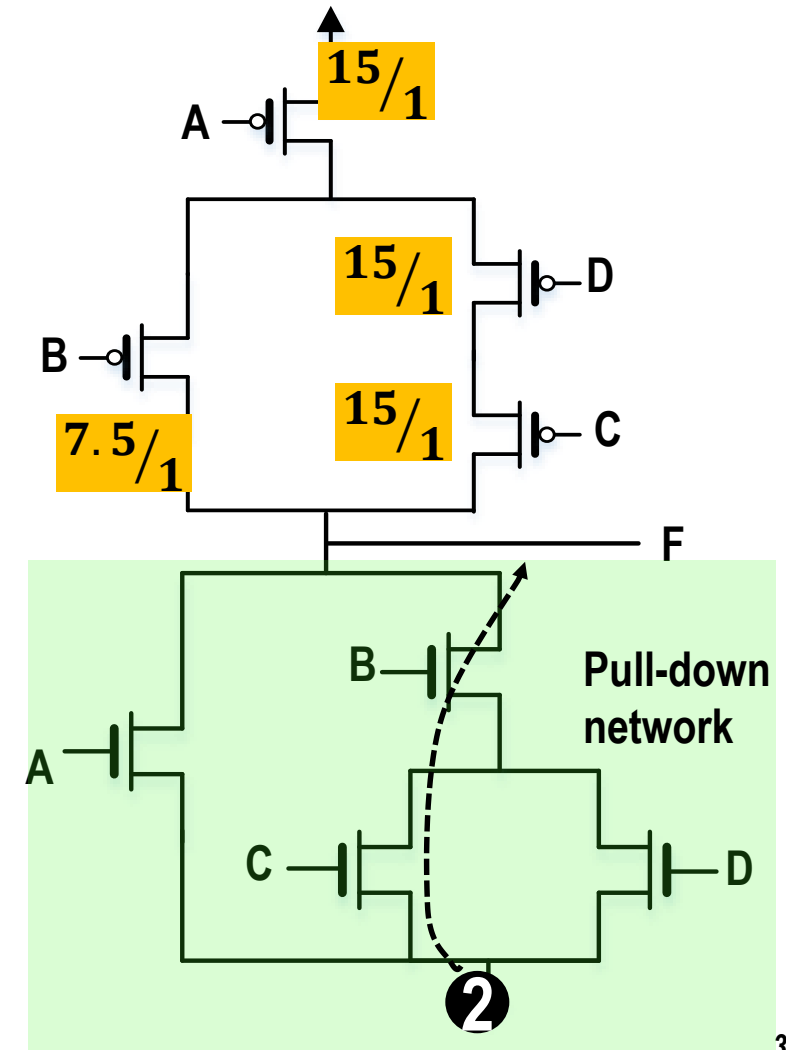
□ For Pull-down network (**Worst Path_2**)

$$\frac{1}{(W/L)_{eq.}} = \frac{1}{(W/L)_B} + \frac{1}{(W/L)_C}$$

$$\frac{1}{2} = \frac{1}{4} + \frac{1}{(W/L)_C}$$

$$\frac{1}{4} = \frac{1}{(W/L)_C}$$

➔ $(W/L)_C = 4/1$



Synthesize and Size of CMOS digital circuit (9/10)



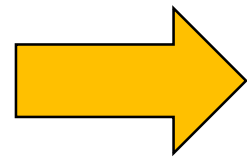
Ex.2. (a) Synthesize the function $F = \overline{A + B(C + D)}$ using CMOS transistors, (b) Sizing the synthesized circuit using a matched design of a reference inverter.

Ans.

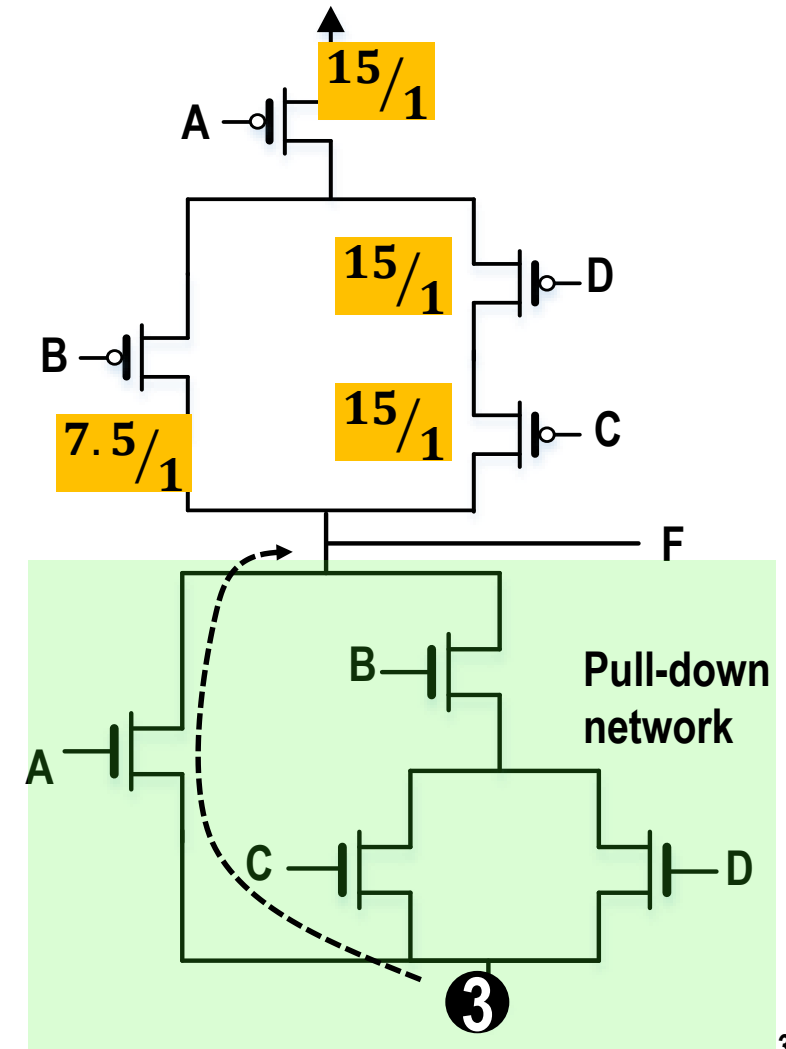
□ For Pull-down network (Path_3)

$$\frac{1}{(W/L)_{eq.}} = \frac{1}{(W/L)_A}$$

$$\frac{1}{2} = \frac{1}{(W/L)_A}$$



$$(W/L)_A = 2/1$$

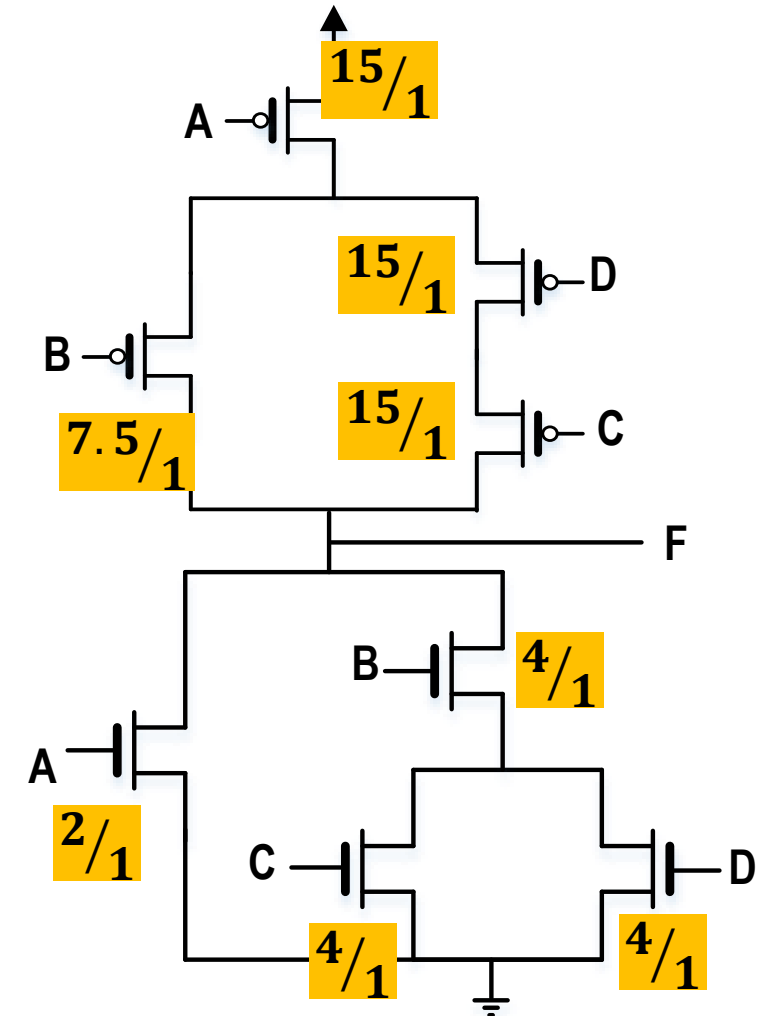
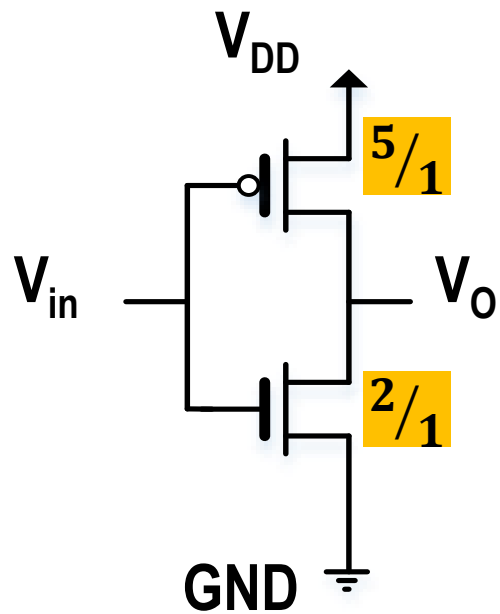


Synthesize and Size of CMOS digital circuit (10/10)



Ex.2. (a) Synthesize the function $F = \overline{A + B(C + D)}$ using CMOS transistors, (b) Sizing the synthesized circuit using a matched design of a reference inverter.

Ans.



Thanks!